Introduction

A quantile is a cut-point that divides a distribution by cumulative probability: the p-quantile is the value below which a proportion p of the data lies. The median is the 0.5-quantile; quartiles are the 0.25, 0.50, and 0.75 quantiles; percentiles are quantiles rescaled to the 0-100 range. Quantiles underlie boxplots, five-number summaries, reference ranges in lab medicine, and non-parametric confidence intervals.

Prerequisites

The reader should know what a cumulative distribution function is and be comfortable with sorted vectors in R.

Theory

For a sample $x_1, \ldots, x_n$ sorted as $x_{(1)} \leq x_{(2)} \leq \ldots \leq x_{(n)}$, the sample p-quantile is not uniquely defined when $p \cdot (n+1)$ is not an integer. R offers nine definitions, selected via the `type` argument of `quantile()`:

- `type = 1`: inverse of the empirical CDF, giving a step-function quantile.
- `type = 4`: linear interpolation of the empirical CDF (SAS default).
- `type = 6`: linear interpolation based on plotting positions (Minitab, SPSS).
- `type = 7`: R's default; linear interpolation between order statistics with no offset.
- `type = 8`: approximately unbiased for continuous distributions (Hyndman-Fan recommendation).

The different types give slightly different numeric values in small samples; they converge as n grows. Type 7 is the default in R's `quantile()`, but reproducibility across software requires specifying the type.

Assumptions

Quantiles are purely descriptive; they require no distributional assumption.

R Implementation

```r
library(dplyr)

set.seed(2026)
x <- rnorm(60, mean = 75, sd = 10)

# Default quantile types
quantile(x, probs = c(0.25, 0.50, 0.75))

# Compare five R types
sapply(c(1, 4, 6, 7, 8), function(t) quantile(x, probs = 0.25, type = t))

# Five-number summary via fivenum
fivenum(x)

# Deciles (10th, 20th, ..., 90th percentiles)
quantile(x, probs = seq(0.1, 0.9, by = 0.1))

# Quantiles by group
diabetes <- tibble::tibble(
  arm = factor(rep(c("Placebo", "Active"), each = 40)),
  hba1c = c(rnorm(40, 7.4, 0.6), rnorm(40, 6.8, 0.7))
)
diabetes |> group_by(arm) |>
```

```r
  summarise(q25 = quantile(hba1c, 0.25),
            q50 = quantile(hba1c, 0.50),
            q75 = quantile(hba1c, 0.75))
```

Output & Results

```
25% 50% 75%
71.2 74.8 80.6

   1    4    6    7    8
72.3 71.8 71.0 71.2 71.3
```

The five types differ by less than 2 units at $n = 60$. For `diabetes`, the active arm's quartiles are shifted lower than placebo's, as expected.

Interpretation

Report quantiles when the distribution is skewed, ordinal, or otherwise not well-summarised by a mean: median with IQR (25th and 75th percentiles) is the standard form. In reference-range contexts (clinical chemistry), the 2.5 and 97.5 percentiles define the central 95% "normal range".

Practical Tips

- Always specify `type` in `quantile()` when reproducing results across software.
- The boxplot uses Type 2 (`fivenum()`) for its hinges; this differs from Type 7 quantiles.
- Quantiles respect monotonic transformations: `quantile(log(x), 0.5) == log(quantile(x, 0.5))` (exactly for the median, approximately for others depending on `type`).
- For very small samples ($n < 10$), quantile estimates are unstable; report the order statistics directly instead.
- Empirical quantiles are biased estimators of population quantiles; use Harrell-Davis (`Hmisc::hdquantile`) for a smoother estimator.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Descriptive statistics and Table 1